

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("Enter command (or exit): ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = 0;

        if(strcmp(command, "exit")==0)
            break;

        pid_t pid = fork();

        if(pid < 0)
        {
            printf("Fork Failed\n");
        }
        else if(pid == 0)
        {
            execlp(command, command, NULL);

            printf("Invalid Command\n");
            exit(1);
        }
    }
}
```

```
alddug@DESKTOP-SCRM05P: ~/OSSP$ tree
.
├── FileManagement
│   ├── b.txt
│   ├── files.txt
│   ├── hello
│   ├── newcopy.txt
│   └── numbers.txt
├── LICENSE
├── Makefile
├── ProcessLock
│   ├── task2
│   └── task2.c
├── README.md
├── assets
│   └── architecture.png
├── bin
├── doc
│   ├── builtin.h
│   ├── design.md
│   ├── executor.h
│   ├── parser.h
│   ├── report.pdf
│   └── shell.h
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
├── structure.txt
└── tests
    ├── test_executor.c
    └── test_parser.c

11 directories, 30 files
alddug@DESKTOP-SCRM05P: ~/OSSP$
```

```
siddu@DESKTOP-SCRM05P:~/ProcessTask$ mkdir ProcessTask
mkdir: cannot create directory 'ProcessTask': File exists
siddu@DESKTOP-SCRM05P:~/ProcessTask$ cd ProcessTask
siddu@DESKTOP-SCRM05P:~/ProcessTask$ nano task2.c
siddu@DESKTOP-SCRM05P:~/ProcessTask$ nano task2.c
siddu@DESKTOP-SCRM05P:~/ProcessTask$ gcc task2.c -o task2
siddu@DESKTOP-SCRM05P:~/ProcessTask$ sudo apt update
[sudo] password for siddu:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.4 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Hit:6 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Fetched 1118 kB in 5s (288 kB/s)
99 packages can be upgraded. Run 'apt list --upgradable' to see them.
gcc is already the newest version (4:15.2.0-5ubuntu1).
gcc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 99
siddu@DESKTOP-SCRM05P:~/ProcessTask$ ./task2
Enter command (or exit): date
Wed Aug  5 09:29:00 UTC 2026
Enter command (or exit): pwd
/home/siddu/ProcessTask
Enter command (or exit): ls
task2  task2.c
Enter command (or exit): whoami
siddu
Enter command (or exit): exit
siddu@DESKTOP-SCRM05P:~/ProcessTask$
```

```

Linux DESKTOP-SCRM5P: ~$ uname -a
Linux DESKTOP-SCRM5P 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
Linux DESKTOP-SCRM5P: ~$ ls
Documents  .vs_lab  file.txt  top_output.txt
First      ProcessTask  lscpu_output.txt  uname_output.txt
First.c    abc.txt  process.c.save  ps_output.txt
CSDP      assignment1_history.txt
Linux DESKTOP-SCRM5P: ~$ ls -l
total 56
drwxr-xr-x 3 shreya shreya 4096 Aug 5 09:02 Documents
-rwxr-xr-x 1 shreya shreya 15768 Aug 4 17:24 First
-rw-r--r-- 1 shreya shreya 23 Aug 4 17:24 First.c
drwxr-xr-x 13 shreya shreya 4096 Aug 4 17:09 CSDP
drwxr-xr-x 2 shreya shreya 4096 Aug 4 18:46 .vs_lab
drwxr-xr-x 2 shreya shreya 4096 Aug 5 09:27 ProcessTask
-rw-r--r-- 1 shreya shreya 0 Aug 4 17:24 abc.txt
-rw-r--r-- 1 shreya shreya 3231 Aug 4 18:34 assignment1_history.txt
-rw-r--r-- 1 shreya shreya 0 Aug 5 09:01 file.txt
-rw-r--r-- 1 shreya shreya 3211 Aug 4 18:50 lscpu_output.txt
-rw-r--r-- 1 shreya shreya 0 Aug 4 17:24 process.c.save
-rw-r--r-- 1 shreya shreya 2869 Aug 4 18:50 ps_output.txt
-rw-r--r-- 1 shreya shreya 3210 Aug 4 18:50 top_output.txt
-rw-r--r-- 1 shreya shreya 125 Aug 4 18:50 uname_output.txt
Linux DESKTOP-SCRM5P: ~$ ls -a
.  ..  .bashrc  .bash_history  .bash_logout  .bash_profile  .bashrc  .cache  .config  .cvs  .emacs  .first  .first.c  .lscpu_output.txt  .process.c.save  .ps_output.txt  .top_output.txt  .uname_output.txt
Linux DESKTOP-SCRM5P: ~$ cd
Linux DESKTOP-SCRM5P: ~$ cd ..
Linux DESKTOP-SCRM5P: ~$ mkdir lab
mkdir: Permission denied
Linux DESKTOP-SCRM5P: ~$ rmdir
rmdir: the following required arguments were not provided:
  <dirs>...

Usage: rmdir [OPTION]... DIRECTORY...

For more information, try '--help'.
Linux DESKTOP-SCRM5P: ~$ rmdir test

```

```
shreyaj@LAPTOP-5T6VIM4D: X Ubuntu X Ubuntu X Ubuntu X + v - X
siddu@DESKTOP-5CRM05P:~$ pwd
/home/siddu
siddu@DESKTOP-5CRM05P:~$ history
 1  sudo apt update
 2  sudo apt install build-essential
 3  gcc --version
 4  gcc --version
 5  pwd
 6  ls
 7  ls
 8  cd OSSP
 9  mkdir OSSP
10  pwd
11  mkdir OSSP
12  cd OSSP
13  pwd
14  mkdir src include obj bin docs tests scripts assets
15  mkdir src include obj bin docs tests scripts assets
16  ls
17  cd src
18  pwd
19  touch main.c shell.c parser.c executor.c builtin.c
20  ls
21  cd ..
22  cd include
23  touch shell.h parser.h executor.h builtin.h
24  ls
25  cd ..
26  cd docs
27  touch shell.h parser.h executor.h builtin.h
28  pwd
29  mkdir OSSP
30  cd OSSP
31  pwd
32  mkdir include obj bin docs tests scripts assets
33  ls
34  cd src
35  touch main.c shell.c parser.c executor.c builtin.c
36  ls
37  cd ..
38  cd include
39  touch shell.h parser.h executor.h builtin.h
40  ls
41  cd ..
42  cd docs
```

```
41 cd ..
42 cd docs
43 touch design.md report.pdf
44 cd ..
45 cd tests
46 touch test_parser.c test_executor.c
47 cd ..
48 cd scripts
49 touch run.sh
50 cd ..
51 cd assets
52 touch architecture.png
53 cd ..
54 touch README.md LICENSE Makefile .gitignore
55 ls -a
56 git init
57 ls -a
58 sudo apt install tree
59 tree
60 tree > structure.txt
61 gcc --version
62 gcc --version
63 git --version
64 cd
65 touch file.txt
66 cd ~/Documents
67 gcc --version
68 git --version
69 cd
70 touch file.txt
71 cd ~/Documents
72 ls ~
73 history
74 gcc --version
75 git --version
76 cd
77 touch file.txt
78 cd ~/OSSP
79 mkdir FileManagement
80 cd FileManagement
81 touch file1.txt
82 ls file1.txt
83 touch a.txt b.txt c.txt
84 echo "Hello Ubuntu"> file1.txt
85 cat file1.txt
```

```
131 cd ProcessTask
132 nano task2.c
133 gcc task2.c -o task2
134 ./task2
135 pwd
136 history
137 whoami
138 date
139 cal
140 hostname
141 uname -a
142 ls
143 cd ~
144 mkdir Documents
145 touch First.c
146 nano First.c
147 gcc First.c -o First
148 touch abc.txt process.c.save
149 ls
150 ls -a
151 ls -l
152 cd
153 cd ..
154 mkdir lab
155 rmdir
156 rmdir test
157 rmdir lab
158 find
159 find . -name "*.OSSP"
160 cd ~
161 history
162 explorer.exe
163 explorer.exe
164 history > assignment1_history.txt
165 explorer.exe
166 history > assignment1_history.txt
167 explorer.exe .
168 pwd
169 ls
170 mkdir OS_LAB
171 cd OS_LAB
172 nano task1.c
173 gcc task1.c -o task1
174 sudo apt update
175 sudo apt install build-essential
```



```
shreyaj@LAPTOP-FT6VM4D: x Ubuntu x Ubuntu x Ubuntu + v x
175 sudo apt install build-essential
176 ./task1
177 uname -a
178 lscpu
179 ps
180 top
181 uname -a
182 lscpu
183 ps
184 top
185 uname -a > uname_output.txt
186 lscpu > lscpu_output.txt
187 ps -ef > ps_output.txt
188 top -b -n 1 > top_output.txt
189 ls
190 history
191 history
192 pwd
193 history
siddu@DESKTOP-5CRM05P:~$ whoami
siddu
siddu@DESKTOP-5CRM05P:~$ date
Wed Aug 5 09:32:58 UTC 2026
siddu@DESKTOP-5CRM05P:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
siddu@DESKTOP-5CRM05P:~$ hostname
DESKTOP-5CRM05P
siddu@DESKTOP-5CRM05P:~$ uname -a
Linux DESKTOP-5CRM05P 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
siddu@DESKTOP-5C
Documents OS_LAB file.txt top_output.txt
First ProcessTask lscpu_output.txt uname_output.txt
First.c abc.txt process.c.save
OSSP assignment1_history.txt ps_output.txt
siddu@DESKTOP-5CRM05P:~$ ls -l
total 56
drwxr-xr-x 3 shreyaj shreyaj 4096 Aug 5 09:02 Documents
-rwxr-xr-x 1 shreyaj shreyaj 15768 Aug 4 17:24 First
-rw-r--r-- 1 shreyaj shreyaj 23 Aug 4 17:24 First.c
drwxr-xr-x 13 shreyaj shreyaj 4096 Aug 4 17:09 OSSP
drwxr-xr-x 2 shreyaj shreyaj 4096 Aug 4 18:46 OS_LAB
drwxr-xr-x 2 shreyaj shreyaj 4096 Aug 5 09:27 ProcessTask
-rw-r--r-- 1 shreyaj shreyaj 0 Aug 4 17:24 abc.txt
-rw-r--r-- 1 shreyaj shreyaj 3231 Aug 4 18:34 assignment1_history.txt
```



```

-rw-r--r-- 1 shreyaj shreyaj    0 Aug  5 09:01 file.txt
-rw-r--r-- 1 shreyaj shreyaj 3211 Aug  4 18:50 lscpu_output.txt
-rw-r--r-- 1 shreyaj shreyaj    0 Aug  4 17:24 process.c.save
-rw-r--r-- 1 shreyaj shreyaj 2869 Aug  4 18:50 ps_output.txt
-rw-r--r-- 1 shreyaj shreyaj 3210 Aug  4 18:50 top_output.txt
-rw-r--r-- 1 shreyaj shreyaj  125 Aug  4 18:50 uname_output.txt
siddu@DESKTOP-5CRM05P:~/home$ ls -a
.          .landscape  First.c          lscpu_output.txt
..         .lesshtst   OSSP             process.c.save
.bash_history .local      OS_LAB           ps_output.txt
.bash_logout .motd_shown ProcessTask       top_output.txt
.bashrc      .profile    abc.txt          uname_output.txt
.cache       Documents   assignment1_history.txt
.config      First      file.txt
siddu@DESKTOP-5CRM05P:~/home$ cd
siddu@DESKTOP-5CRM05P:~$ cd ..
siddu@DESKTOP-5CRM05P:~/$ mkdir lab
mkdir: Permission denied
siddu@DESKTOP-5CRM05P:~/$ rmdir
error: the following required arguments were not provided:
  <dirs>...

Usage: rmdir [OPTION]... DIRECTORY...

For more information, try '--help'.
siddu@DESKTOP-5CRM05P:~$ cd ..
siddu@DESKTOP-5CRM05P:~/$ rmdir folder_name
rmdir: failed to remove 'folder_name': No such file or directory
siddu@DESKTOP-5CRM05P:~/$ rmdir test
rmdir: failed to remove 'test': No such file or directory
siddu@DESKTOP-5CRM05P:~/$ rmdir lab
siddu@DESKTOP-5CRM05P:~/$ find
./assignment1_history.txt
./OS_LAB
./OS_LAB/task1
./OS_LAB/task1.c
./.local
./.local/share
./.local/share/nano
./process.c.save
./ps_output.txt
./.motd_shown
./OSSP

```

```
./OSSP/docs
./OSSP/docs/design.md
./OSSP/docs/report.pdf
./OSSP/docs/shell.h
./OSSP/docs/parser.h
./OSSP/docs/builtin.h
./OSSP/docs/executor.h
./OSSP/assets
./OSSP/assets/architecture.png
./OSSP/Makefile
./OSSP/LICENSE
./OSSP/FileManager
./OSSP/FileManager/b.txt
./OSSP/FileManager/newcopy.txt
./OSSP/FileManager/numbers.txt
./OSSP/FileManager/hello
./OSSP/FileManager/file1.txt
./OSSP/.gitignore
./OSSP/tests
./OSSP/tests/test_parser.c
./OSSP/tests/test_executor.c
./OSSP/structure.txt
./OSSP/src
./OSSP/src/main.c
./OSSP/src/shell.c
./OSSP/src/parser.c
./OSSP/src/builtin.c
./OSSP/src/executor.c
./OSSP/scripts
./OSSP/scripts/run.sh
./OSSP/README.md
./OSSP/include
./OSSP/include/shell.h
./OSSP/include/parser.h
./OSSP/include/builtin.h
./OSSP/include/executor.h
./OSSP/ProcessTask
./OSSP/ProcessTask/task2.c
./OSSP/ProcessTask/task2
./OSSP/bin
./OSSP/obj
./OSSP/.git
./OSSP/.git/hooks
./OSSP/.git/hooks/prepare-commit-msg.sample
./OSSP/.git/hooks/post-update.sample
```

```
./OSSP/.git/info
./OSSP/.git/info/exclude
./OSSP/.git/HEAD
./OSSP/.git/objects
./OSSP/.git/objects/info
./OSSP/.git/objects/pack
./OSSP/.git/config
./OSSP/.git/description
./OSSP/.git/refs
./OSSP/.git/refs/tags
./OSSP/.git/refs/heads
./top_output.txt
./First
./abc.txt
./bash_history
./ProcessTask
./ProcessTask/task2.c
./ProcessTask/task2
./Documents
./Documents/FileManagement
./Documents/FileManagement/b.txt
./Documents/FileManagement/newcopy.txt
./Documents/FileManagement/numbers.txt
./Documents/FileManagement/hello
./Documents/FileManagement/file1.txt
./bashrc
./lscpu_output.txt
./landscape
./landscape/sysinfo.log
./First.c
./bash_logout
./config
./config/ubuntu-insights
./config/ubuntu-insights/linux-consent.toml
./config/ubuntu-insights/wsl_setup-consent.toml
./config/ubuntu-insights/ubuntu_release_upgrader-consent.toml
./config/procps
./cache
./cache/ubuntu-insights
./cache/ubuntu-insights/wsl_setup
./cache/ubuntu-insights/wsl_setup/local
./cache/ubuntu-insights/wsl_setup/uploaded
./cache/ubuntu-insights/wsl_setup/uploaded/1785130289.json
./cache/ubuntu-insights/linux
./cache/ubuntu-insights/linux/local
```